



ECE 639
Selected Topics in Electronics
Power Management
Spring 2021

Topic 3
Low-Dropout Regulators (LDOs)

Sameh A. Ibrahim
Ain Shams University
ICL

**(Courtesy of E. Sinencio – Texas A&M, M. El-Nozahi – Ain Shams University
and F. Elseddeek – Cairo University)**

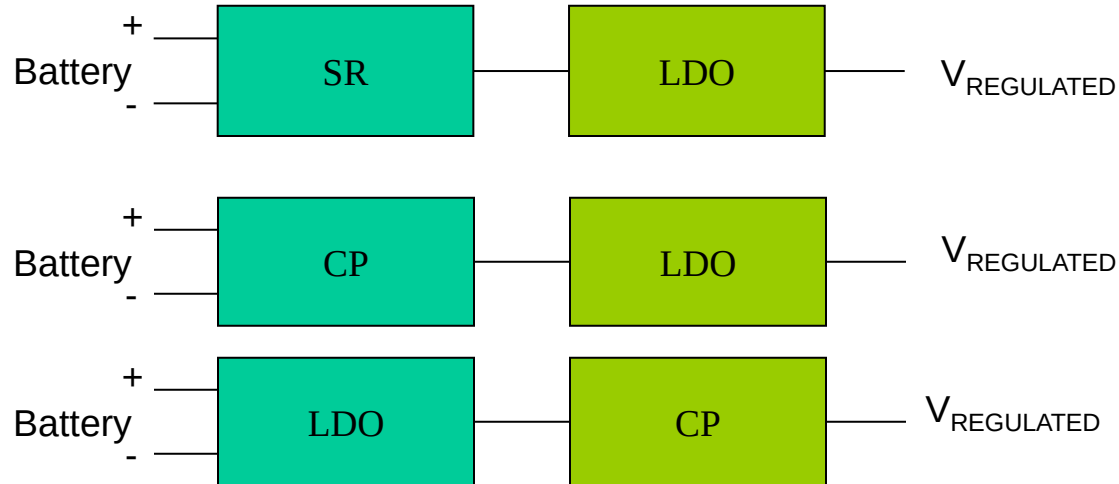
Outline

- **Linear Regulators Principles**
- **Low-Dropout Regulators (LDOs) Circuits**
- **LDO Parameters**
- **Towards a better LDO**
 - Capacitor-less
 - High PSR
 - Fast Transients
 - Low IQ
 - Low Noise
 - Digital LDO



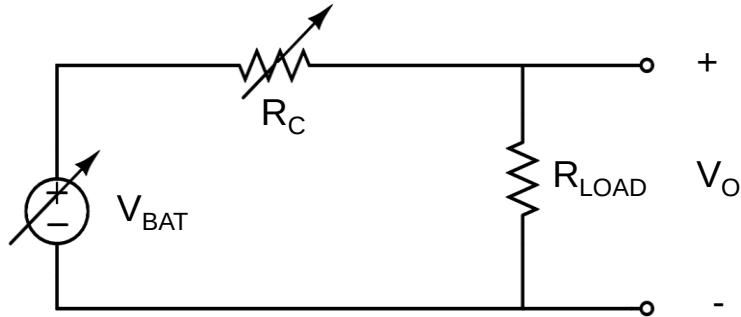
Power Conversion

- Objective of a power converter is to provide a regulated output voltage



	LDO	SR	CP
Efficiency	Not controllable	Up to 95%	Moderate
Integration	Depends on the architecture	Requires external components	Depends on the architecture
Output Ripples	No	Yes	Yes

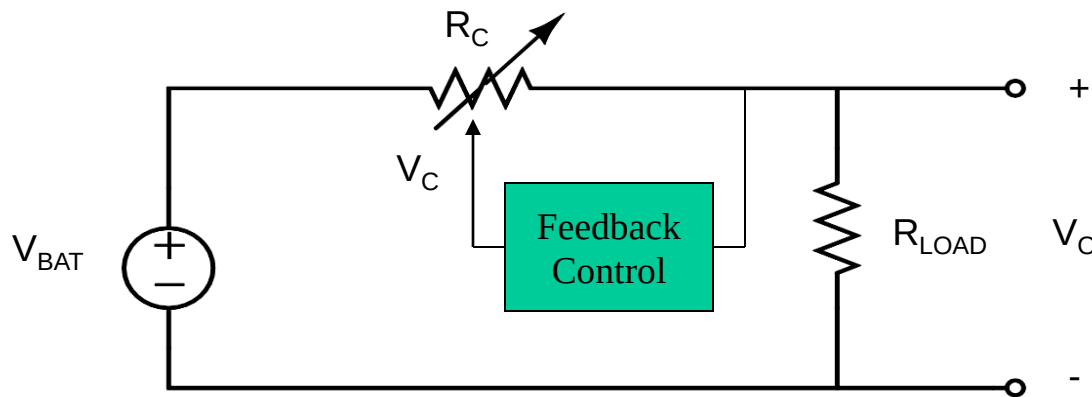
Linear Regulator: Principles



- V_O must be constant and $R_C \ll R_{LOAD}$
- V_{BAT} is changing as a function of time

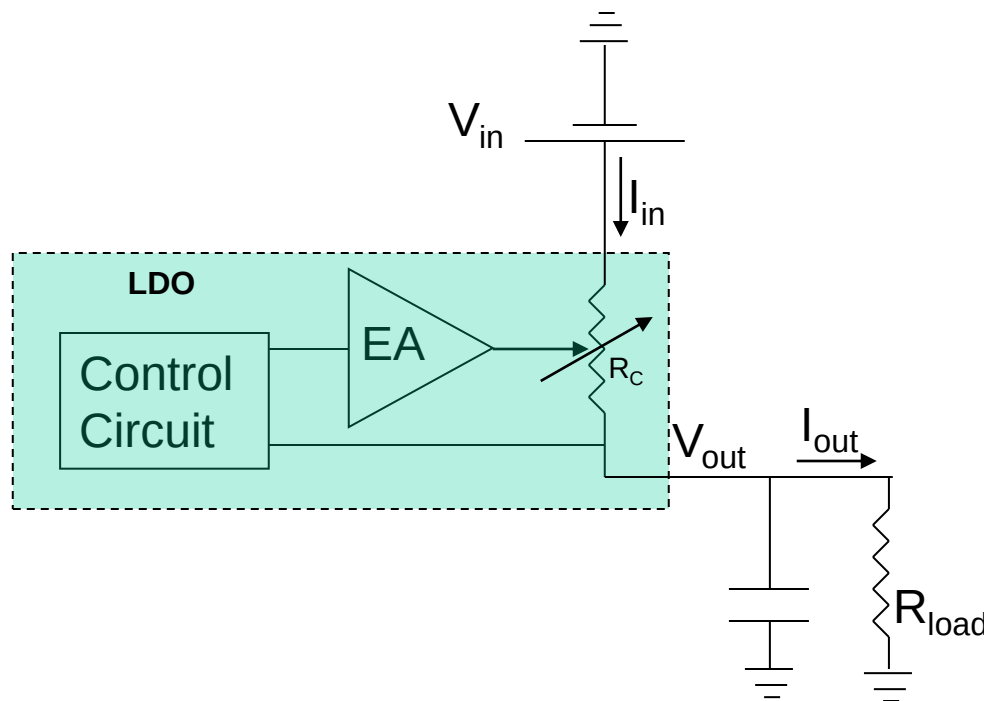
$$V_O = \frac{R_{LOAD}}{R_{LOAD} + R_C} V_{BAT}$$

Thus in order to keep constant V_O , the value of the controlling resistor R_C should adapt based on the battery voltage.



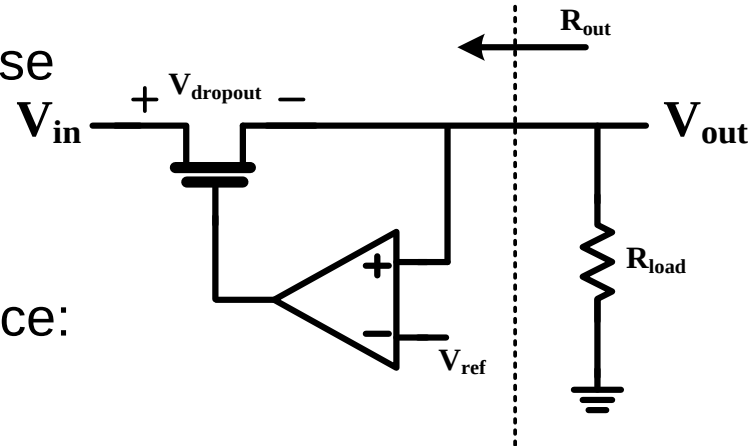
Low-Dropout Voltage Regulator (LDO)

- The LDO acts as a variable resistor that is placed between input power source and the load in order to drop and control the voltage applied to the load.



LDO FEATURES

- No switching involved, no switching noise
- Very quiet output
- $V_{in} > V_{out}$ to allow for current flow.
- V_{in} and V_{out} have the same polarity.
- Low R_{out} to act as an ideal voltage source:
 - Constant output voltage
 - AC ground behavior



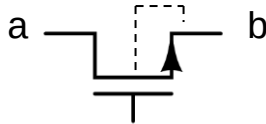
- Low dropout voltage, to maximize V_{in} dynamic range and improve efficiency.

$$\eta = \frac{P_{out}}{P_{in}} = \frac{I_o V_{out}}{I_{in} V_{in}} = \frac{I_o V_{out}}{(I_q + I_o)(V_{dropout} + V_{out})} \leq \frac{V_{out}}{V_{dropout} + V_{out}} \quad (\text{for high } I_o)$$

- Efficiency at high load current, is determined by $V_{dropout}$.
- Efficiency at low load current is determined by I_q and $V_{dropout}$.
- Low power LDOs: Load current $< 1A$ (portable devices).
- High power LDOs: Load current $> 1A$ (automotive & industrial devices).

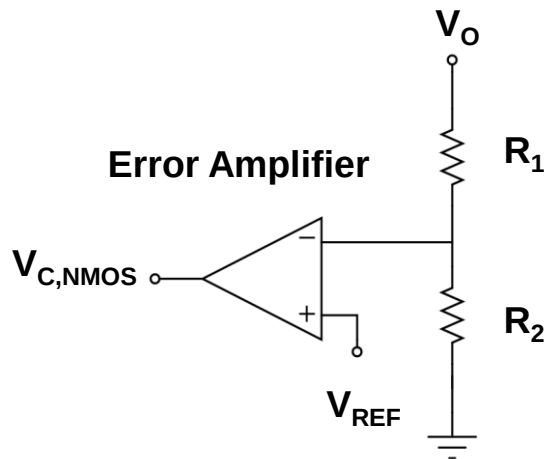
Implementing R_C and the Feedback Control

NMOS Pass Transistor

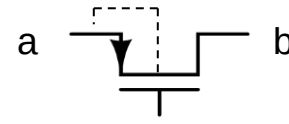


$$V_C = V_{out} + V_{GS} = V_{in} - V_{dsat,opamp}$$

$$V_{DO,n} = V_{dsat,opamp} + V_{GS}$$

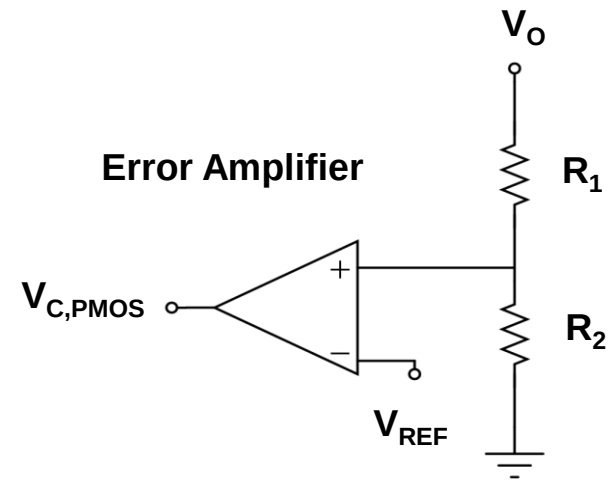


PMOS Pass Transistor



$$V_C = V_{in} - V_{SG}$$

$$V_{DO,p} = V_{SD(SAT)}$$



- V_{REF} is a Bandgap voltage which is also supplied by $V_{BAT} = V_{IN}$.
- NMOS requires a charge pump for low dropout.

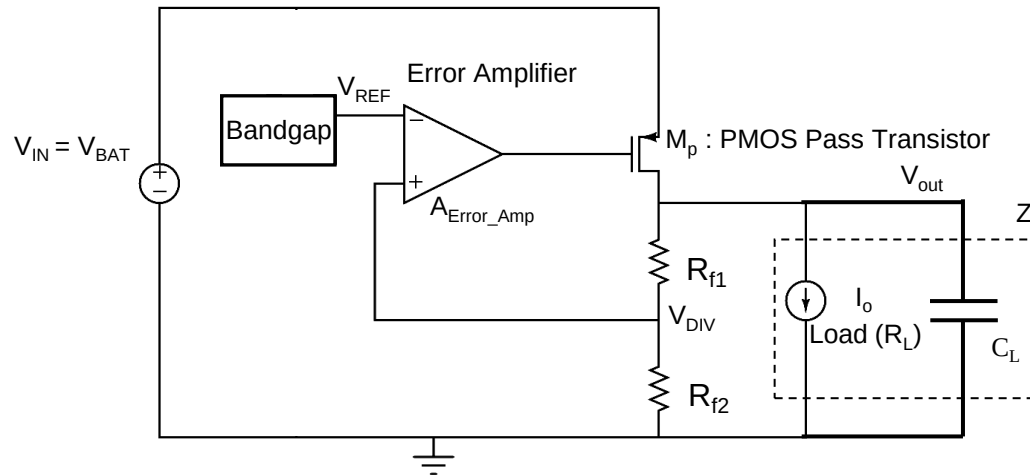


NMOS vs PMOS

- NMOS pass FET is easier to compensate at low loads and dropout, due to the higher output impedance of PMOS.
- NMOS pass FET are smaller due to weaker drive of PMOS.
- NMOS pass FET LDO requires the VDD rail to be higher than V_{in} , while a PMOS does not. To do this, a charge pump is usually required with accompanying disadvantages of higher quiescent current and extra charge pump noise.
- Power Supply Rejection (PSR) is better with PMOS as output is at the drain of the transistor.



V_{out} in terms of V_{REF}

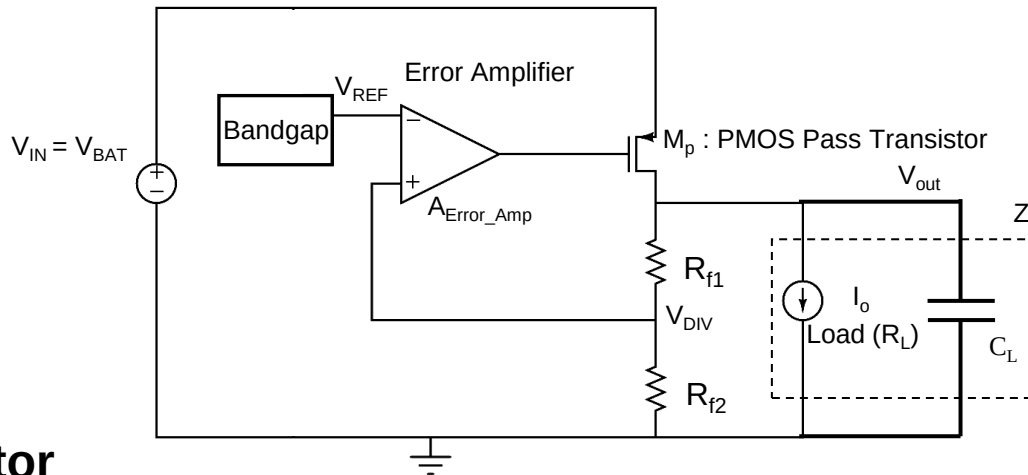


$$V_{out} = V_{REF} \left(\frac{A_{open_loop}}{1 + A_{open_loop} \left(\frac{R_{f2}}{R_{f1} + R_{f2}} \right)} \right)$$

- Because of the negative feedback, the output voltage is a function of V_{REF} .



Issues of Concern in LDO Design



- **Pass transistor**

- Load current will determine its size and thus layout

- **Error amplifier**

- The accuracy required by the LDO, determines the magnitude of the open loop gain.
- Single pole architectures are recommended for better and easier stability.
- The amp transient requirement is dependent on the stability i.e. gain and phase margins. There is a trade-off in making the PM high and speed of amp. This is also true for the Gain.
- Should have high PSRR

- **Bandgap voltage reference**

- Should have high PSR

- **Stability and speed of the feedback loop**

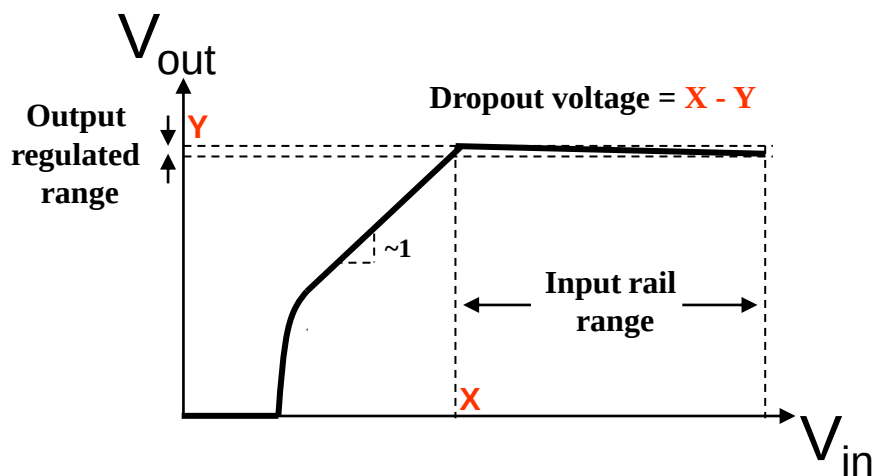
- Should be assured under all load conditions

- **Choice of the capacitors and feedback resistors (R_{f1} and R_{f2})**

Low Dropout Regulators

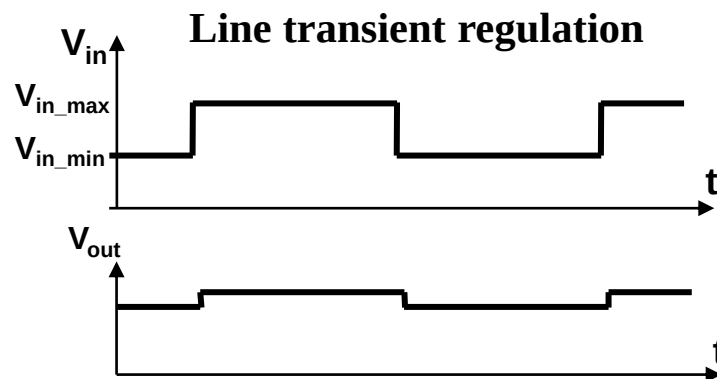
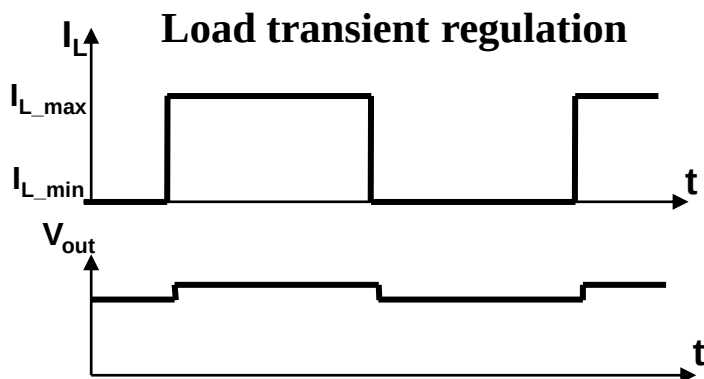
LDO Significant Parameters (1)

- **Dropout voltage (V_{do}):** This is the difference between the minimum voltage the input DC supply can attain and the regulated output voltage.
- **Input rail range:** This is the input supply voltage range that can be regulated. Lower limit is dependent on the dropout voltage and upper limit depends on the process capability.
- **Output regulated voltage range:** This is the output voltage variation the regulator guarantees. When output voltage is in this range, it is said to be in regulation.



LDO Significant Parameters (2)

- **Output current range:** This is the output current handling capability of the regulated output voltage.
 - Minimum current limit is mainly dependent on the stability requirements
 - Maximum current limit is dependent on Safe Operating Area (SOA) of pass FET and also maintaining output voltage in regulation.
- **Load regulation:** This is the variation in output voltage as current moves from min. to max.
- **Line regulation:** This is the variation in output voltage as supply voltage is varied from min. to max..
- **Load/Line transient regulation:** This is a measure of the response speed of the regulator when subjected to a fast load/Vsupply change.



LDO Significant Parameters (3)

- **PSR:** Power Supply Rejection (or ripple rejection) is a measure of the ac coupling between the input supply voltage on the output voltage.
- **Power Efficiency:** This is the ratio of the output load power consumption to input supply power. Linear regulators are not really efficient especially at high input supply voltages.
$$\eta \equiv \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT} I_{LOAD}}{V_{IN} I_{IN}} \approx \frac{V_{OUT}}{V_{IN}}$$
- **Output capacitor range:** This is the specified output capacitance the regulator is expected to accommodate without going unstable for a given load current range.
- **Overshoot/Undershoot:** It is important to minimized high transient voltages at start-up and during load and line transients.



Outline

- Linear Regulators Principles
- Low-Dropout Regulators (LDOs) Circuits
- LDO Parameters
- **Towards a better LDO**
 - Capacitor-less
 - High PSR
 - Fast Transients
 - Low IQ
 - Low Noise
 - Digital LDO

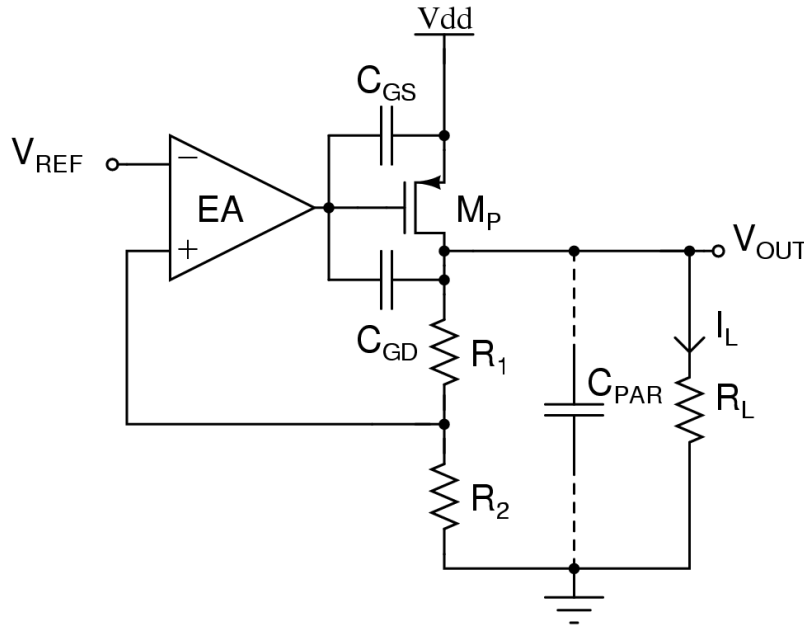


Capless LDOs

- Traditionally, LDOs had a large capacitor at the output ($> 100 \text{ nF}$).
- This guarantees LDO stability.
- This is still OK for stand-alone LDOs or for very-low output ripples.
- SOCs require (multiple) integrated LDOs.
- No extra pins are available for output capacitors.
- Capless LDOs are thus needed.
- Stability is an issue.



Stability (Two Gain Stages)



Poles Location

$$\omega_{EA} = \frac{1}{R_{O,EA} (C_{GS} + (1 + g_{m,MP} (R_{ds} \parallel (R_1 + R_2) \parallel R_L)) C_{GD})}$$

$$\omega_{OUT} = \frac{1}{C_{PAR} (R_{ds} \parallel (R_1 + R_2) \parallel R_L)}$$

$$C_{PAR} = C_{out} + C_{parasitic}$$

$$C_{parasitic} : C_{GD} + C_{BD} + \dots$$

$$C_{OUT} : \text{Compensation Cap}$$

- Consider that poles are located at the output of the error amplifier (EA) and the output of the LDO.
- If EA pole is the dominant pole, thus the output pole need to be placed beyond unity gain frequency (UGF) to achieve stability

- Note that poles are function of load current (I_L)

$$\omega_{EA} \propto \sqrt{I_L}$$

$$\omega_{OUT} \propto I_L$$

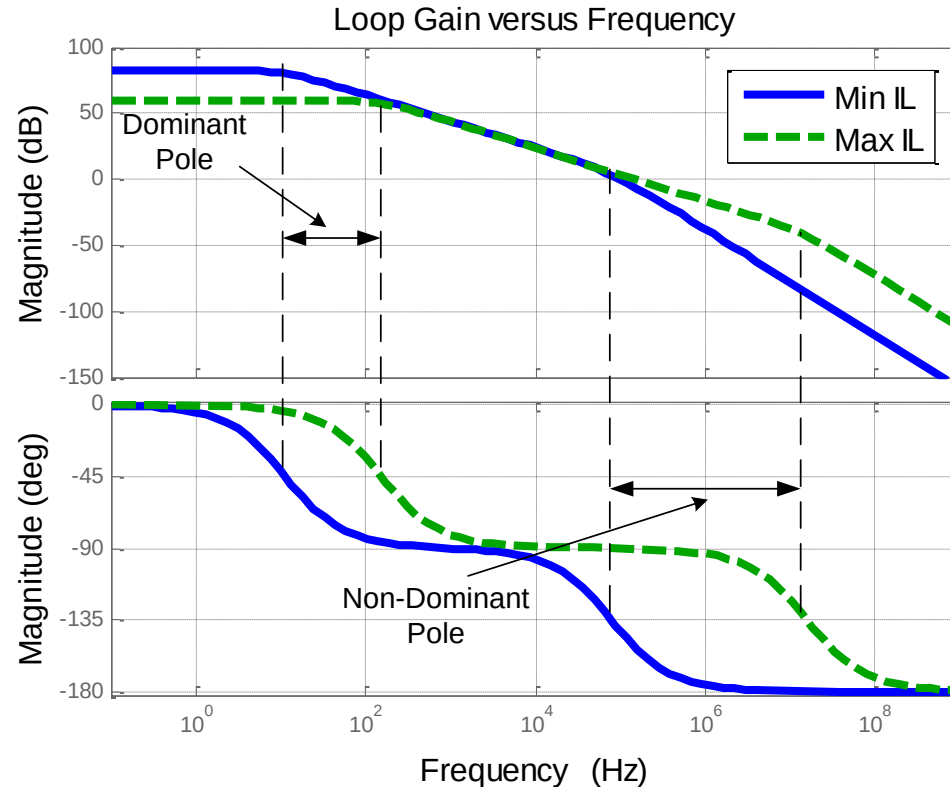
$$\omega_{OUT} \gg \omega_{EA} \quad \text{For Capless LDOs}$$

$$\omega_{OUT} \ll \omega_{EA} \quad \text{For LDOs with External } (C_L) \text{ Capacitor}$$

Low Dropout Regulators

Poles Movement (ω_{EA} & ω_{OUT})

Assuming Capless



- Phase margin at light loads (Min I_L) is reduced.
- Observe that output pole frequency reduces at a much faster rate than the EA pole frequency with reducing load, thus creating stability issues at light loads.



Miller-Compensated Capless LDO

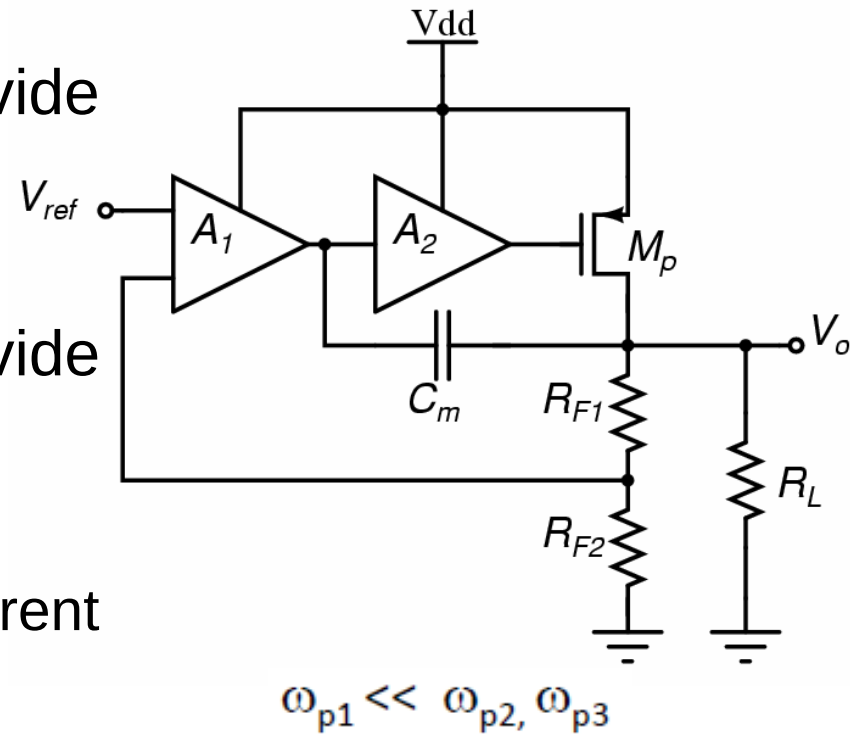
- **Main Features**

- Basic architecture to provide regulation

- **Principle of Operation**

- Amplifier A_1 is used to provide the necessary regulation.
- Amplifier A_2 serves for:
 - Providing the necessary current to drive the pass transistor.
 - Amplifying the Miller capacitance (C_m) putting the dominant pole at the output of A_1 .

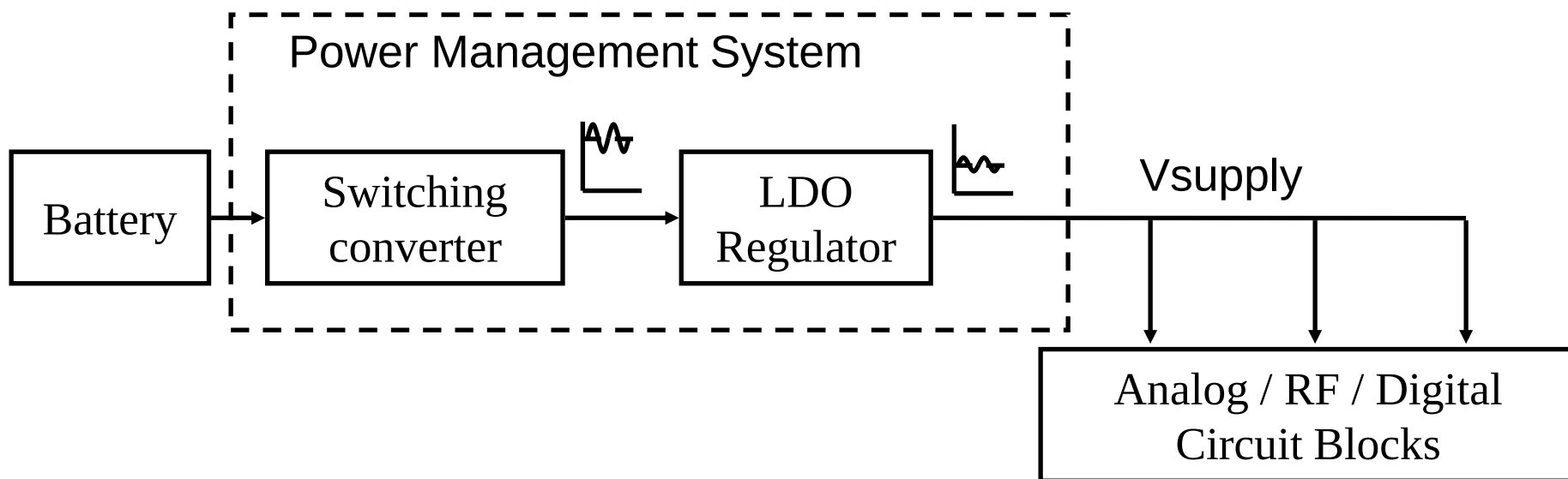
- Dominant pole is determined through the Miller capacitor.



$\omega_{p1} \ll \omega_{p2}, \omega_{p3}$
[Leung, JSSC 2003]



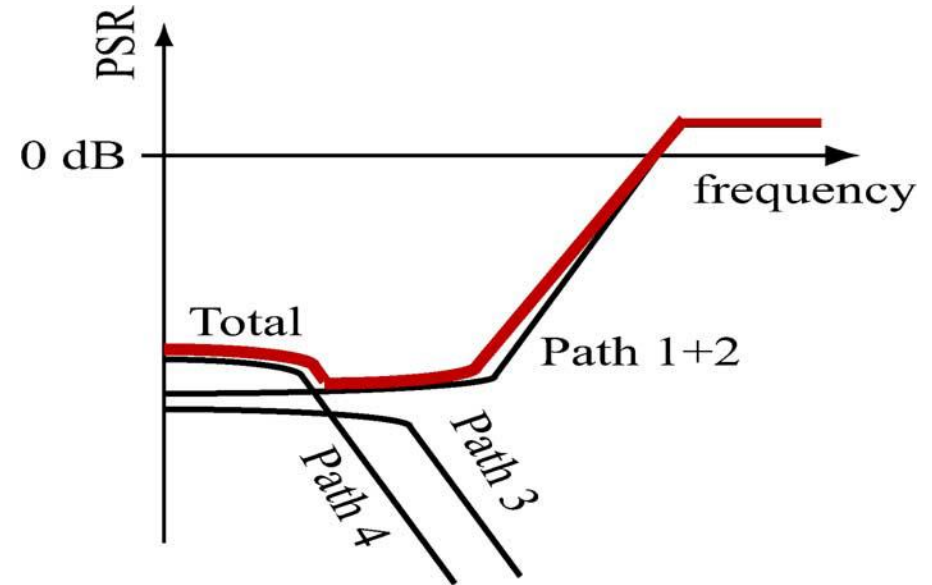
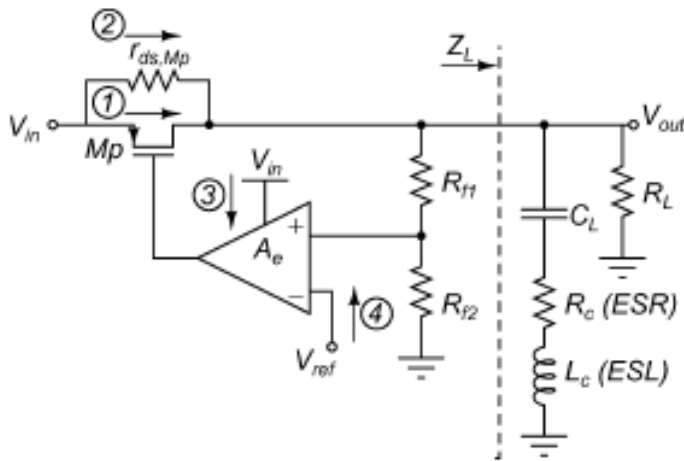
Power Supply Rejection



- **Problem:**
 - Supply ripples affect Analog/RF blocks.
 - Switching converter ripple frequencies are increasing.
- **Solution: LDO with good PSR at higher operating frequencies**
- **Challenges: Low drop-out voltage, low quiescent current, small area, high PSR across a wide frequency range**



Main Supply Leakage Paths



- **Input-Output Ripples Paths**

- Low frequency: Bandgap (V_{REF}) and Error Amplifier
- High frequency: Pass transistor and loop GBW



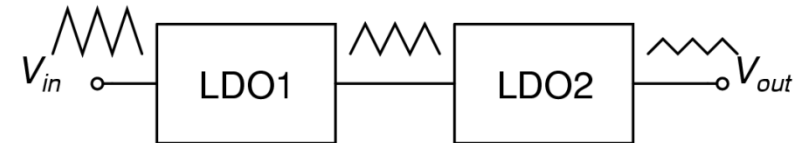
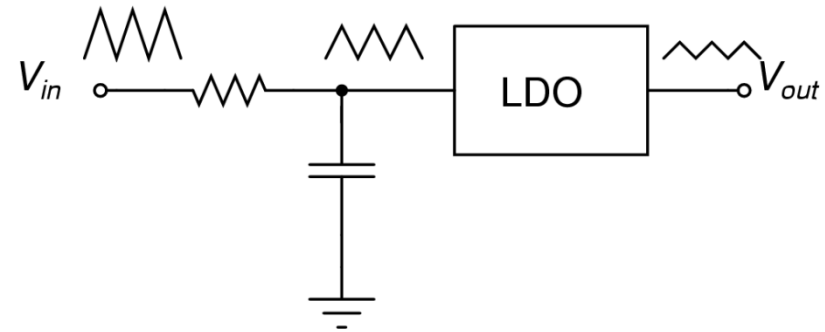
Existing Techniques for better PSR

- **Existing Techniques:**

- RC filtering
- Cascading LDOs
- Combined RC and cascading
- Increasing Loop Bandwidth

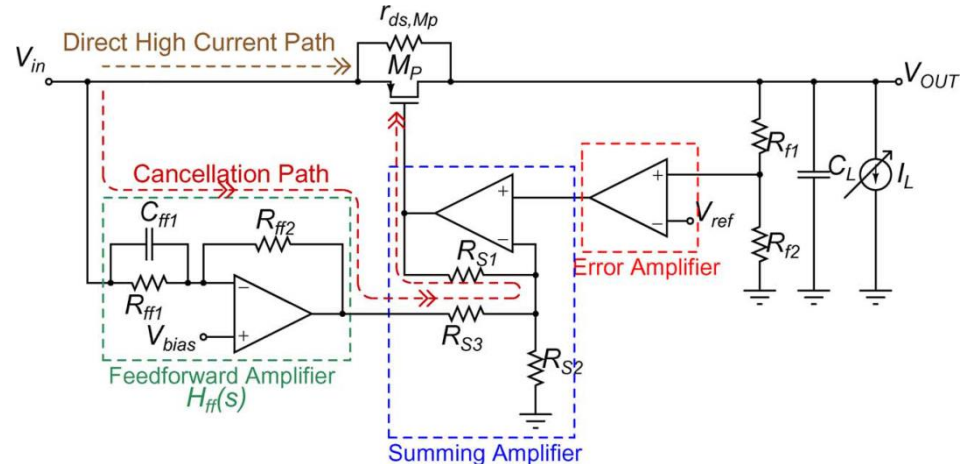
- **Drawbacks:**

- Large area consumption
- Large dropout voltage
- High power consumption



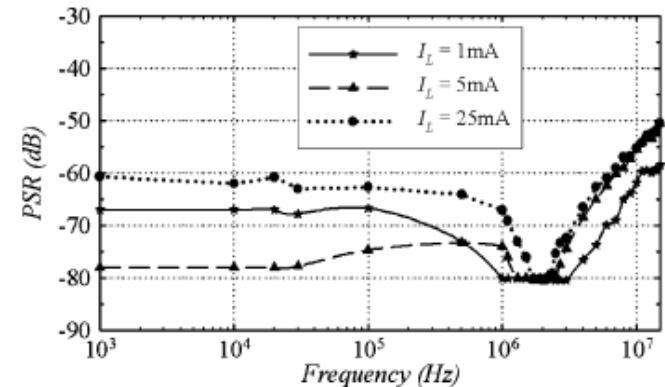
High Frequency PSRR Improvement (FF)

- FF path couples ripples to the gate to be subtracted from ripples on source.
- FF amplifier cannot perfectly equalize the phase delay at all frequencies and operation conditions.
- Summing amplifier is needed complicating the design.

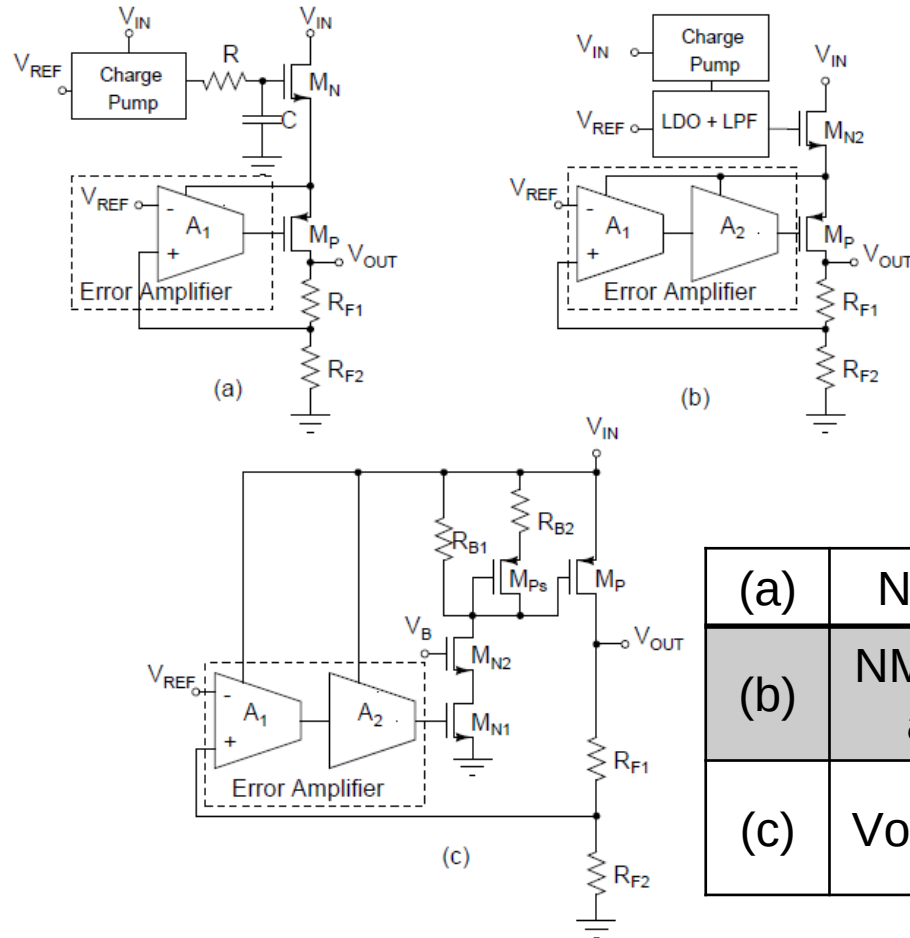


$$H_{ff}(s)|_{opt} = \frac{1 + \frac{s}{\omega_s}}{A_{SQ}} \cdot \left(1 + \frac{1}{g_{m,Mp} r_{ds,Mp}} \right)$$

$$= \frac{1 + \frac{s}{\omega_s}}{A_{SQ}} \cdot \left(\frac{g_{m,Mp} + g_{ds,Mp}}{g_{m,Mp}} \right).$$

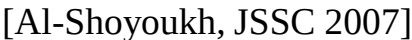


Capless High-PSR LDOs



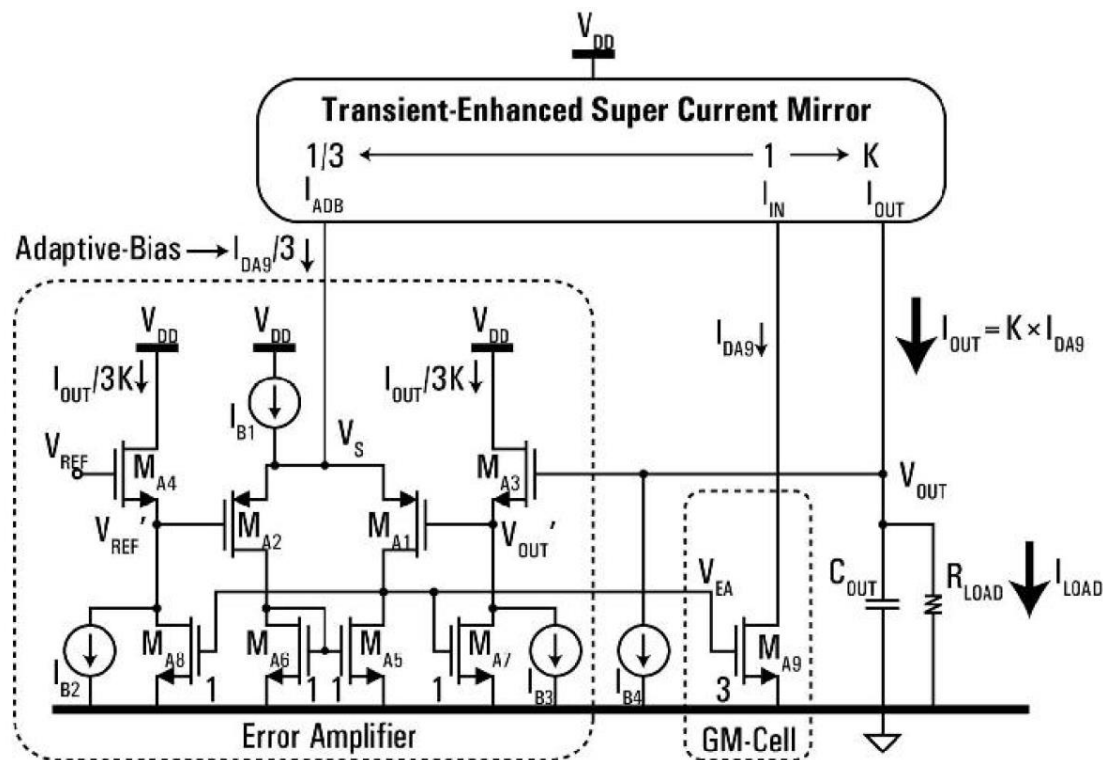
(a)	NMOS Cascode	[Gupta JSSC 2007]
(b)	NMOS Cascode & auxiliary LDO	[Zhan APCCAS 2010]
(c)	Voltage Subtractor	[Yang WSEASTCAS 2008]

[Torres, CAS Mag. 2014]



- ## Low Dropout Regulators

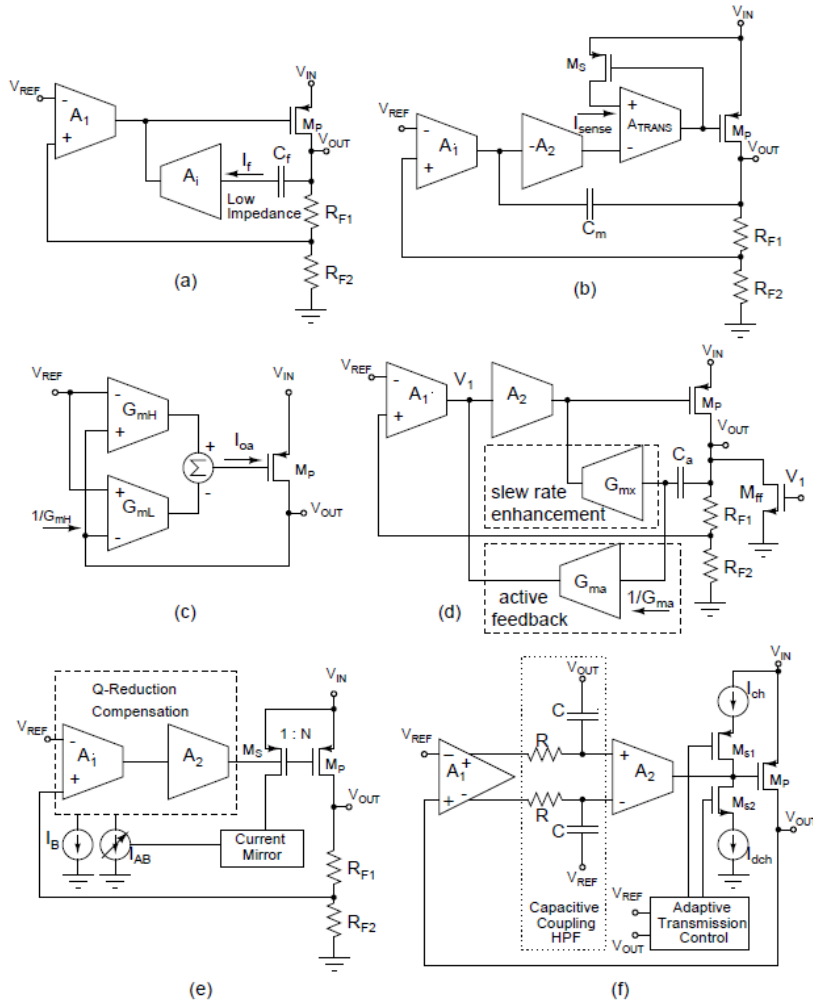
Fast Transient LDOs (2)



[LAM, ISSCC 2008]

- A solution for capless LDOs.
- Capless LDOs suffer more from load transients.
- Adaptive biasing provide more current when needed.

Other Fast-Transient Capless LDOs



(a)	Current Amplifier	[Milliken TCASI 2007]
(b)	Current sense transistor and TIA	[Chen ASSCC 2007]
(c)	High slew-rate EA	[MAN TCASII 2007]
(d)	Active FB compensation and SR Enhancement	[HO TCAS II 2010]
(e)	Adaptive Bias	[ZHAN TCASI 2010]
(f)	CC HPF & ATC	[ZHEN TCASI 2011]

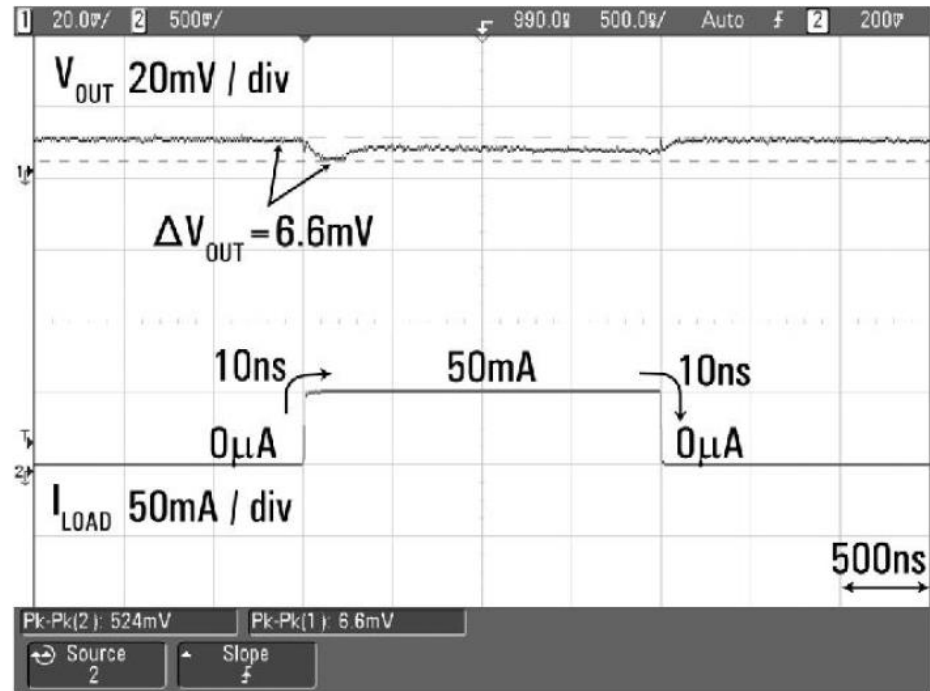
[Torres, CAS Mag. 2014]

Transient Response FOM

$$FOM(ns) = \frac{t_r I_q}{I_{L,max}}$$

$$t_r = \frac{C_L \Delta V_{out}}{I_{L,max}}$$

$$FOM(ns) = \frac{C_L \Delta V_{out} I_q}{I_{L,max}^2}$$

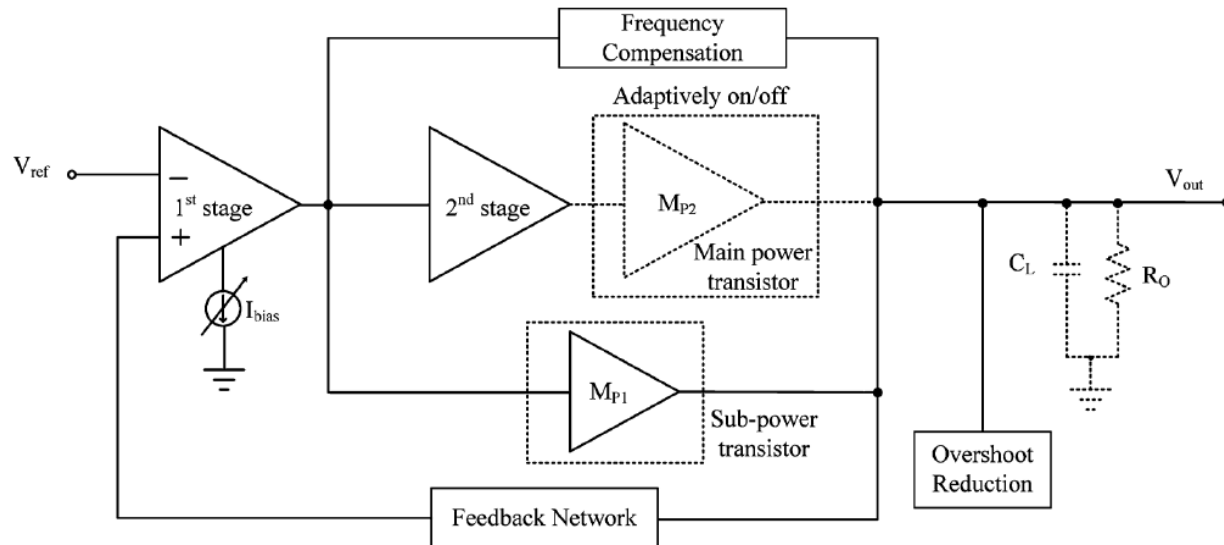


Low-IQ LDOs

- The key in reducing I_q of an LDO is reducing the I_q of the BGR and the EA.
- Ultra-low power LDOs are important for energy harvesting and wearable applications.
- Ultra-low power BGRs and EAs suffer from performance degradation.
- Other auxiliary loops are thus needed to improve the performance needed.



A Capless Ultra-Low-Iq LDO



[CHONG, TCAS I 2013]

- **0.9- μ A I_q**
- **Adaptively turning ON/OFF pass transistor and EA stages based on load**
- **Dynamic Biasing techniques for EA**

Noise in an LDO

$$\overline{V_{n,out}^2} = \overline{V_{R_1}^2} + \overline{V_{R_2}^2} + \overline{V_{ref}^2} + \overline{V_{EA}^2} + \overline{V_{mos}^2} + \overline{V_{extcap}^2}$$

$$\overline{V_{R_1}^2} = \int_{f_1}^{f_2} 4kTR_1 df$$

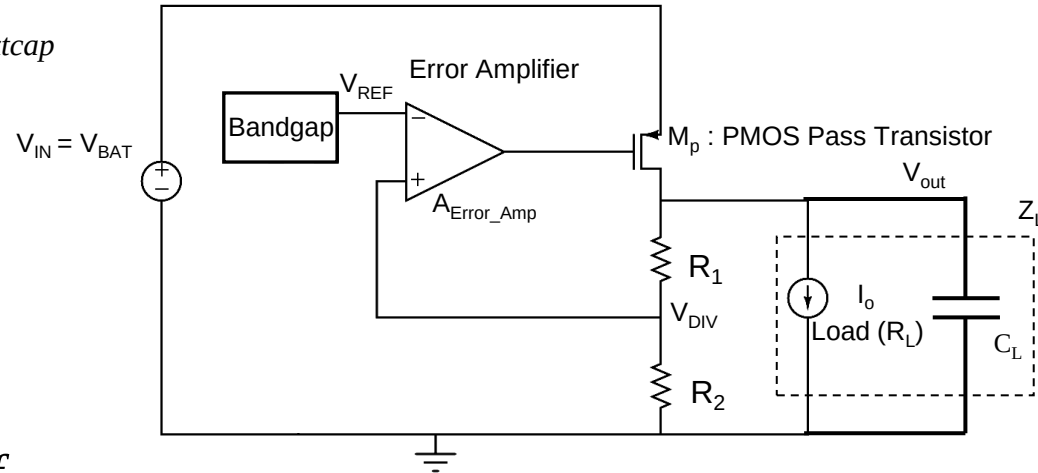
$$\overline{V_{R_2}^2} = \int_{f_1}^{f_2} 4kTR_2 \left(\frac{1}{\beta} \right)^2 df, \beta = \frac{R_2}{R_1 + R_2}$$

$$\overline{V_{ref}^2} + \overline{V_{EA}^2} = \int_{f_1}^{f_2} \left(\overline{V_{n,BG}^2} + \overline{V_{nin,EA}^2} \right) \left(\frac{1}{\beta} \right)^2 df$$

$$\overline{V_{mos}^2} = \int_{f_1}^{f_2} 4kT \left[\frac{2}{3} g_m (R_1 \parallel r_{ds} \parallel R_L)^2 \times \frac{1}{1 + (R_1 \parallel r_{ds} \parallel R_L)^2 C_L^2 (2\pi f)^2} \right] df$$

$$+ \int_{f_1}^{f_2} \frac{K_p g_m^2 (R_1 \parallel r_{ds} \parallel R_L)^2}{C_{ox} W L f} \times \frac{1}{1 + (R_1 \parallel r_{ds} \parallel R_L)^2 C_L^2 (2\pi f)^2} df$$

$$\overline{V_{extcap}^2} = \frac{kT}{C_L}$$



[Monteiro, JLPE 2009]

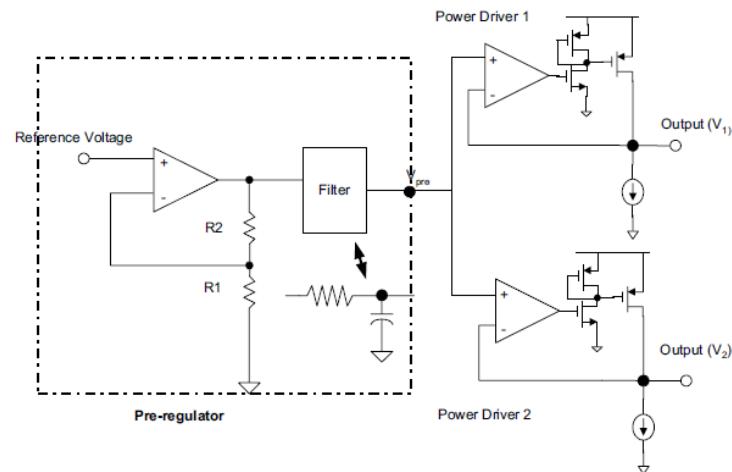
Low noise requires larger β , larger C_L , and low-noise BG and EA.

Low Dropout Regulators



Example of a Low-Noise LDO

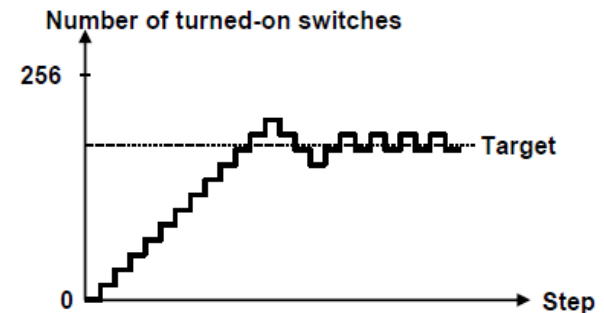
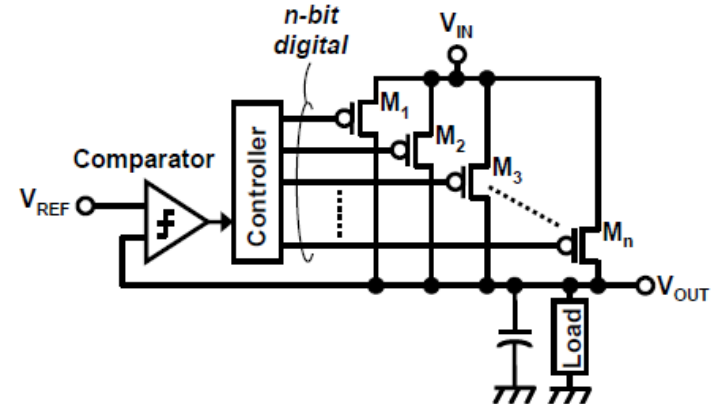
- Cascade of LDOs
- A filter to minimize noise of first LDO
- $\beta=1$ for second LDO
- Dual output
- Diode-connected stage to improve PSR



[Hoon, CICC 2005]

Digital LDO

- Error amplifier replaced by a comparator
- Regulation via number of active pass transistors rather than gate voltage
- Allows lower input voltage.
 - Paper reports 0.5-V V_{in}
- Suffers from inherent output ripples even for constant loads due to finite resolution of switch sizes.
- A system clock is needed.
- Digital controller determines the speed of the LDO.



[Okuma, JSSC 2010]

